

Matthew Pinto

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Experience

Research Assistant | University of Maryland College Park

(July 2026 - Present)

- Programming Python scripts using NumPy to process 3D medical image volumes and manipulate voxel arrays.
- Implementing 3D dilation and spatial transformation algorithms to model vascular geometry changes in target Regions of Interest (ROI).
- Evaluating function robustness by testing boundary conditions, voxel-overwrite behavior, and spatial alignment across multi-layer tissue geometries.

GxP Systems Engineering Intern | Millipore Sigma

(June 2026 – August 2026)

- Developed PowerShell automation that reduced manual audit trail review length by over 90% increasing quality of review, improving inspection readiness and speeding up reviewer throughput
- Standardized log outputs and data structures across enterprise asset management tools to ensure full traceability and GxP compliance
- Built and evaluated machine learning models to identify drivers of protracted audit-trail length; performed structured data collection, preprocessing, and feature analysis.
- Generated synthetic audit trail datasets to test and validate PowerShell automation logic, ensuring consistent behavior across edge cases.
- Conceived and implemented a custom subsequence-parsing strategy to replace traditional delimiter formats (tab/comma), successfully resolving complex multi-line log parsing challenges.
- Collaborated cross-functionally with QA, Validation, and Systems Engineering teams to ensure automated outputs were clear, verifiable, and fully aligned with data-integrity protocols
- Applied Lean Six Sigma DMAIC principles; earned White Belt and gained Yellow Belt-level experience through operational-excellence projects.
- Practiced GxP-compliant system handling, including secure access management, audit-trail integrity, and adherence to safety and data security protocols.

Research Assistant | Syracuse University

(November 2024 – July 2026)

- Conducted research on biomaterial scaffolds for cartilage tissue engineering, focused on enhancing mechanical resilience and cellular compatibility.
- Synthesized and optimized biopolymer-based cryogels designed for injectable or load-bearing applications in regenerative medicine.
- Utilized Python, Excel, and statistical tests including ANOVA, post-hoc Tukey, linear regression, and correlation for data analysis.
- Utilized compression testing, rheometry, and degradation analysis to evaluate mechanical and viscoelastic behavior.
- Performed cell seeding, viability assays, and fluorescence microscopy to assess biological response and matrix infiltration.
- Operated instrumentation including mechanical testers, freeze dryers, incubators, and fluorescence microscopes to support data collection and scaffold characterization.
- Presented progress reports and contributed to data interpretation and experimental planning within a multidisciplinary research team.

Education

University of Maryland College Park (Universities at Shady Grove Campus)

Coursework toward Bachelor of Science in Biocomputational Engineering

Graduating May 21, 2027)

Syracuse University

Coursework toward Bachelor of Science in Biomedical Engineering

(Aug 2023 – May 2025)

Publications

Zhang, K., Seitz, M. P., Pinto, M., Eghan, W. O.-A., & Jain, E. (2026). Ice Templated PEG-Alginate Double-Network Cryogels with Tunable Mechanics and Degradation for Soft Tissue Engineering. *Gels*, 12(6), 533.

<https://doi.org/10.3390/gels1206053>

Leadership

- **Vice President of Financials:** AΦE International Fraternity, Syracuse University (Nov 2024 – June 2025)
- **Awards:** Dean's List (Fall 2023 – Spring 2026), 2024 Invent@SU Competition (2nd Place), ChABSA Scholarship (Fall 2025 – Spring 2027), Invest in SUccess Scholarship (Spring 2024 – Spring 2025), 2025 Lambda Phi Epsilon Academic Excellence Award

